VERSION 1.0

— SOURCING & INTEGRATION · COMPREHENSIVE REPORT

Sourcing & Integration of Color E-Paper Displays

From panel to system — an integrated decision & engineering guide

Two tracks: **Indoor** & **Outdoor** · Procurement tiers: **Complete kit / Bare panel / Driver board**

Integration with **Raspberry Pi · ESP32/Arduino/STM32 · Android**

Verified specs, prices & sourcing channels — tailored for Doha's climate

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Benchmark: **Samsung Color E-Paper (EM32DX)**

Sources: **E Ink · Good Display · Geniatech**
Waveshare · IRIS Optronics · buyepaper · Samsung

COMPREHENSIVE REPORT · VERSION 1.0

Contents

SECTION 01

Executive Summary & Decision At-a-Glance

01

This report consolidates a multi-stage investigation into sourcing and integrating **color E-Paper displays** as an alternative to the Samsung Color E-Paper — in **panel + board**, **bare panel**, or **standalone driver board** form — for two separate use cases: **indoor** and **outdoor** in Doha's climate. All figures and specifications below are drawn from official product pages and specialist sources.

Two decisions govern everything else: **(a) temperature** dictates the technology (indoor ≠ outdoor), and **(b) the panel's interface** dictates the driver board (large color panels require a matching TCON — there is no generic board).

TRACK	RECOMMENDED OPTION	TEMP	SUGGESTED CONTROLLER
Indoor	**Good Display 31.5" Spectra 6 + ESP32-S3 board**	0 → 50°C	ESP32 (embedded), or Android/RPi via WiFi/USB.
Outdoor	**E Ink Kaleido 3 Outdoor 25.3" + Luvia board**	−15 → 65°C	Official E Ink board + IP65 enclosure & shading.

THE FOUR KEY POINTS: ① Do not use a Spectra 6 panel (indoor, 0–50°C) outdoors in Doha. ② A large color panel (25.3"/31.5") requires the manufacturer's TCON — the generic IT8951 is grayscale only. ③ Android does not drive SPI directly; its role is a content source over WiFi/USB to the driver board. ④ **Sequence the purchase (§11):** nothing ≥25.3" is bought until a one-page content spec is written and at least two supplier quotes are in hand — the pipeline is proven first on a ~$298 13.3" kit.

SECTION 02

Governing Principles

02

2.1 Principle one: temperature separates the two tracks

- **Spectra 6** (most vivid color, 6-particle): operates **0–50°C** → indoor. Doha summers reach 45–50°C ambient, and a panel surface in direct sun hits 60–70°C → risk of ghosting and permanent damage outdoors.
- **Kaleido 3 Outdoor**: **−15 to 65°C** → practical outdoor, sunlight-readable, solar-capable (muted color, color-filter technology).
- **Marquee**: **−20 to 65°C**, vivid outdoor color, but 55" and partner-only.
- **ChLCD** (IRIS Optronics): **−20 to 70°C**, the most robust color in direct sun, but a B2B solution.
- **"Indoor" is not always <50°C in Qatar:** glass atriums, full-height-glazed lobbies, and window-adjacent retail can reach 55–60°C in direct sun — beyond Spectra 6's range. Treat such sun-exposed indoor spots as outdoor for panel selection.

2.2 Principle two: the panel↔driver pairing rule

A panel is never wired directly; it terminates in a fragile FPC ribbon that plugs into an adapter/driver board, which then connects to the host. The driver board type is set by the panel's interface:

- **Small SPI (≤13.3"):** a cheap adapter (DESPI / Waveshare HAT) + your own controller (RPi/ESP32/STM32). Flexible and economical.
- **Large color (25.3" Mini-LVDS / 31.5" QSPI 60-pin):** the manufacturer's matching TCON only — no generic board exists.
- **Large grayscale Carta:** the generic IT8951 (USB/SPI/I80) — but black & white, no color.

> **Practical rule for Doha**
>
> Even outdoor-rated panels (−15/65°C) need an IP65 enclosure + sun-shade + anti-glare, because direct sun can push the panel surface above 65°C in summer. Refresh also slows in high heat.

2.3 Design goals & non-goals (from the project technical design document)

The kiosk project's TDD fixes the requirements that every selection in this report must serve — and, just as importantly, what the system deliberately does *not* attempt:

GOALS

- **G1 Environmental resilience** — reliable operation across −20→+60°C ambient, IP65 for outdoor sites.
- **G2 Offline-first** — full function with no network: local rendering and caching.
- **G3 Power efficiency** — zero standby power on the display subsystem (bistability).
- **G4 Procurement traceability** — structured RFQ→deployment workflow with audit trail (§11.5).
- **G5 Content integrity** — CRC validation + staleness detection on every push (§7.5).
- **G6 Prototype-first** — pipeline proven on a low-cost kit before any production panel is ordered (§11.2).

NON-GOALS

- **NG1** No video or animation — e-paper refresh is inherently slow; designing for it invites failure.
- **NG2** No cloud dependency — the system must be complete offline; cloud is an attack surface and a liability, not a feature.
- **NG3** No multi-vendor panel mixing within one deployment — one panel type per site keeps waveforms, spares, and procedures uniform.
- **NG4** No blanket color — color panels only where a genuine stakeholder requirement exists (the §11.1 decision rule).

SECTION 03

Color E-Paper Technology Comparison

03

TECHNOLOGY	COLOR	TEMP	AVAILABLE AS PANEL+BOARD	BEST FOR
Spectra 6	6-particle — most vivid (60,000-color gamut)	0→50°C	Readily available (Good Display)	Indoor color
Kaleido 3 Outdoor	4096 (color-filter) — muted	−15→65°C	Available (+ Luvia/CONCERTO)	Practical outdoor color
ChLCD (IRIS)	Accurate, vivid color	−20→70°C	B2B / contact	Harsh outdoor (direct sun)
Marquee	4-particle — vivid	−20→65°C	Partner-only (55")	Large vivid outdoor
Carta	Black & white (16 grays)	Wide	Available (+ IT8951)	Economical outdoor, no color

Note: Spectra 6 won SID's Display of the Year award; E Ink is the world's largest ePaper supplier (sources: E Ink, SID). Temperature figures are from official product pages.

SECTION 04

Track 1 — Indoor Use (color)

04

All use **E Ink Spectra 6** (the most vivid), at **0W** static and **under 6W** during updates. The closest size to the Samsung 32" is **31.5"**.

PRODUCT	SIZE / RESOLUTION	BOARD / INTERFACE	APPROX. PRICE	NOTE
Good Display GDEP315C01(E6)	31.5" / 2560×1440	+ ESP32-S3 board (WiFi/BT/TF)	~$700–1,200	Closest to Samsung. QSPI 60-pin. PMIC.
Good Display GDEP253C02(E6)	25.3" / Mini-LVDS	+ TCON + Android board (USB)	~$1,000–1,400	Easy USB updates (copy images).
Good Display GDEP133C02(E6)	13.3" / 1600×1200	+ ESP32-133C02 (WiFi/BLE/USB/SD)	~$300–449	Built-in web server. Ideal for prototyping.
E Ink Online Shop (direct)	25.3" / 13.3" / 7.3"	+ separate driving board	$1,400 / $449 / $149	31.5" currently out of stock there.
Waveshare	up to 13.3"	+ ESP32-S3 / HAT	4": $50 · 13.3": ~$298–327	Cheapest; does not exceed 13.3".

RECOMMENDATION: **Good Display 31.5" Spectra 6 + ESP32-S3** — the direct equivalent of the Samsung display. For prototyping/budget: Waveshare 13.3" (~$298). For a direct purchase with factory warranty: E Ink's shop (25.3" at $1,400).

Sources: *good-display.com, buyepaper.com, shopkits.eink.com, Waveshare.*

Hot-indoor caveat (Qatar)

Spectra 6's 0–50°C range leaves no margin in sun-exposed interiors — glass atriums, glazed lobbies, west-facing windows — where Doha indoor temperatures can reach 55–60°C. For those locations choose an outdoor-rated technology (Kaleido 3 Outdoor / ChLCD) or a wide-temp B&W panel, not Spectra 6.

SECTION 05

Track 2 — Outdoor Use (Qatar Heat)

05

Color options are narrower here. The practical panel+board choice is **Kaleido 3 Outdoor**; for the most robust color in sun, **ChLCD**; for the largest/most vivid via partners, **Marquee**.

PRODUCT / TECHNOLOGY	SIZE	FORM / BOARD	TEMP	NOTE
E Ink Kaleido 3 Outdoor	25.3"	+ Luvia board (separate)	−15→65°C	Best in practice. Muted color (filter). Solar.
E Ink Kaleido 3 Outdoor	13.3"	+ CONCERTO board (panel ~$449)	−15→65°C	For smaller outdoor prototypes.
Geniatech EPC2530 (Kaleido 3)	25.3"	Semi-finished unit (CMS, 4G, battery)	Outdoor	3200×1800 native; effective color res lower (color-filter). Sunlight-readable, remote mgmt.
IRIS Optronics — ChLCD	Signage	B2B solution (no standalone board)	−20→70°C	Most robust, accurate color in sun + IP56.
E Ink Marquee (4-particle)	55"	Partner-only	−20→65°C	Large vivid color; not panel+board; new.
Wide-temp B&W 31.2" (GDEP312TT3 / DMPH312EC1 / E Ink V32-OSM)	31.2"	Bare panel, or IP65 outdoor unit	−15→65°C	2560×1440, 16-gray. Robust & low-cost, but B&W (no color).
SEEKINK (S315E6 / S280EC-O)	28"–31.5"	Billboard / outdoor-rated	incl. −15→65°C	Alternative supplier: Spectra 6 billboards + outdoor models.

RECOMMENDATION: **Kaleido 3 Outdoor 25.3" + Luvia board** (balance of color/temp/availability). For more accurate, robust color → **ChLCD**. Large & vivid via partners → **Marquee**. Color not required → **wide-temp B&W 31.2" (GDEP312TT3, or the IP65 V32-OSM / DMPH312EC1)**.

Sources: *shopkits.eink.com (Kaleido Outdoor + Luvia/CONCERTO; V32-OSM), eink.com, good-display.com (GDEP312TT3/DMPH312EC1), geniatech.com, iris-opt.com, seekink.com.*

SECTION 06

The Three Procurement Tiers

06

6.1 Complete kit (panel + board)

Fastest to deploy: panel + matching driver board + code & docs. This is the practical route for large color panels (see the two track tables). Example: Good Display 31.5" Spectra 6 + the ready ESP32-S3 board with WiFi/BT/TF.

> **Why choose a complete kit?** The manufacturer has already solved the hardest integration problem: matching the panel's specific waveform requirements to the TCON's output. You get working firmware, example code, and a warranty that covers the pair.

6.2 Bare panel (panel only) — when does it make sense?

- **Small (≤13.3" SPI):** sensible and economical — bare panel + a DESPI adapter + your own controller. Example bare-panel prices: Waveshare 4" ~$49.52, 13.3" ~$326; E Ink: 7.3" ~$149, 13.3" ~$449.
- **Large color (25.3"/31.5"):** "panel only" is not a real saving — it still needs the manufacturer's TCON (no generic board); suitable only if you are designing your own driver board (PMIC + waveforms).
- Caution: the FPC ribbon is fragile (bend only along the horizontal), and the panel runs at 3.3V, so a level shifter is needed with a 5V Arduino.

6.3 Standalone driver boards — catalog by category

CATEGORY / EXAMPLE	DRIVES	HOST CONNECTION	NOTE
DESPI adapters (C02/C73/C1085...)	Small SPI panels ≤13.3"	SPI (GPIO) + your MCU	Cheap ~$5–20; DESPI-L includes USB serial.
ESP32 dev boards (ESP32-133C02, 31.5" board, Waveshare E6)	Specific Spectra 6 color panels	WiFi/BT/USB	The practical color route ~$30–80; web server.
E Ink official (CONCERTO, Luvia)	Kaleido 3 (Outdoor) modules	Per design	Design support from E Ink; price on request.
IT8951 (generic)	Carta grayscale 6"–13.3"+	USB / SPI / I80	Black & white only (16 grays) ~$40–70.

CATEGORY / EXAMPLE	DRIVES	HOST CONNECTION	NOTE
Waveshare HAT (40-pin)	Waveshare SPI panels	Raspberry Pi GPIO	Direct mounting on RPi.

Sources: *good-display.com (DESPI/ESP32-133C02), shopkits.eink.com (CONCERTO/Luvia), waveshare.com (IT8951/HAT).*

SECTION 07

Integration & Wiring (Hosts)

07

The panels work with **Raspberry Pi**, **ESP32/Arduino/STM32**, and **Android** — but the role and connection method differ:

7.1 Raspberry Pi (best compatibility)

- Small/medium SPI (≤13.3" color): via a **HAT/DESPI** on the 40-pin header. Spectra 6 panels connect to RPi/Jetson/Arduino/STM32 through a driver HAT with a standard 40-pin header.
- Large grayscale (Carta): via **IT8951** over USB (or SPI/I80).

7.2 Arduino / ESP32 / STM32 (embedded control)

- SPI connection via **DESPI + FPC cable**; with STM32/ESP32/Arduino code from Good Display.
- **ESP32 is effectively the standard** — the color driver boards themselves are built on it (ESP32-133C02 for the 13.3" color: QSPI + WiFi/BLE/USB/SD).
- Caution: an Arduino Uno is memory-constrained for large color panels (large framebuffer) — use ESP32/STM32.

7.3 Android (the source — fits the kiosk)

- Android does not typically drive SPI directly (restricted GPIO access). Its role is a **content source** sending images to the driver board (the actual controller) via:
- **USB:** fast updates for Android devices, or copy images to a USB drive and process automatically.
- **WiFi:** a built-in web server you update from any phone/device on the same network.
- Flow: Android app → WiFi/USB → TCON board → panel.

7.4 Connection points (the cable / custom part)

- Panel → flat **FPC** ribbon → adapter connector: lift the latch, insert the panel (white side up), close the latch.
- Small SPI: the **DESPI** series. Large color: the connector built into the matching TCON (e.g., 60-pin for the 13.3" QSPI).
- Then driver board → host via **SPI** (GPIO), **USB** (IT8951/Android), or **WiFi** (ESP32 boards).
- Caution: the FPC ribbon is fragile (bend only horizontally); 3.3V logic (level shifter for 5V).

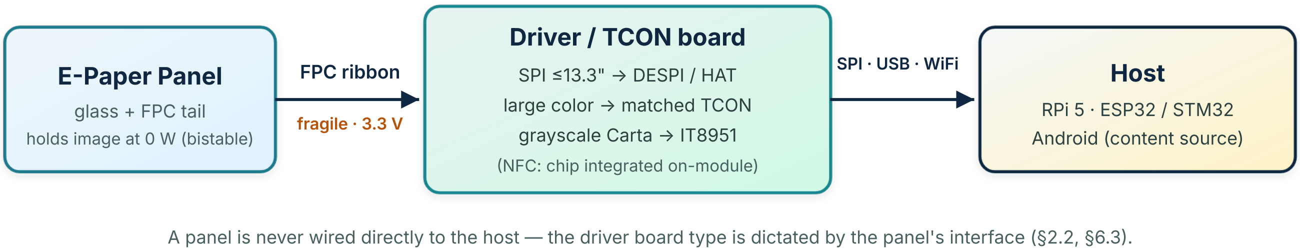


Figure 1 — Physical signal chain: panel → FPC → driver board → host.

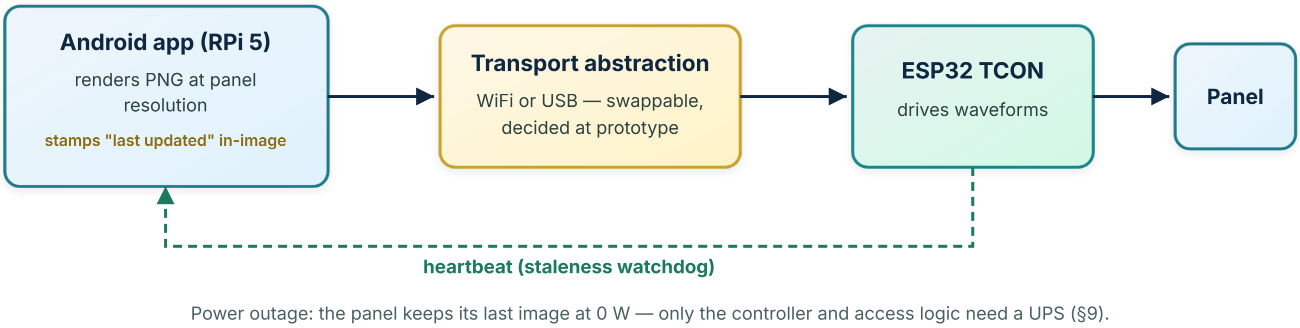


Figure 2 — Kiosk content-push architecture: content logic separated from panel driving, hardened with the staleness watchdog.

Proposed architecture for the kiosk project (RPi5 + Android + ESP32)

Make the **driver board (ESP32 TCON or E Ink board) the direct controller of the panel**, fed by the **RPi5/Android as a content source over WiFi/USB**. This separates **content logic** (RPi/Android) from **panel driving** (TCON) — the same critical/non-critical separation principle you applied in the kiosk design. The result: reliable updates, easier maintenance, and fault isolation. Two hardening rules complete the architecture: **(a) formalize the interface** — render to a fixed intermediate (PNG at panel resolution) and push through a thin transport abstraction, so WiFi vs USB stays a deferred, cheap choice and the panel remains swappable without touching app logic; **(b) add a staleness watchdog** — bistability means a wedged TCON keeps showing yesterday's content convincingly, so the TCON must emit a heartbeat and the app must render a "last updated" timestamp into every pushed image.

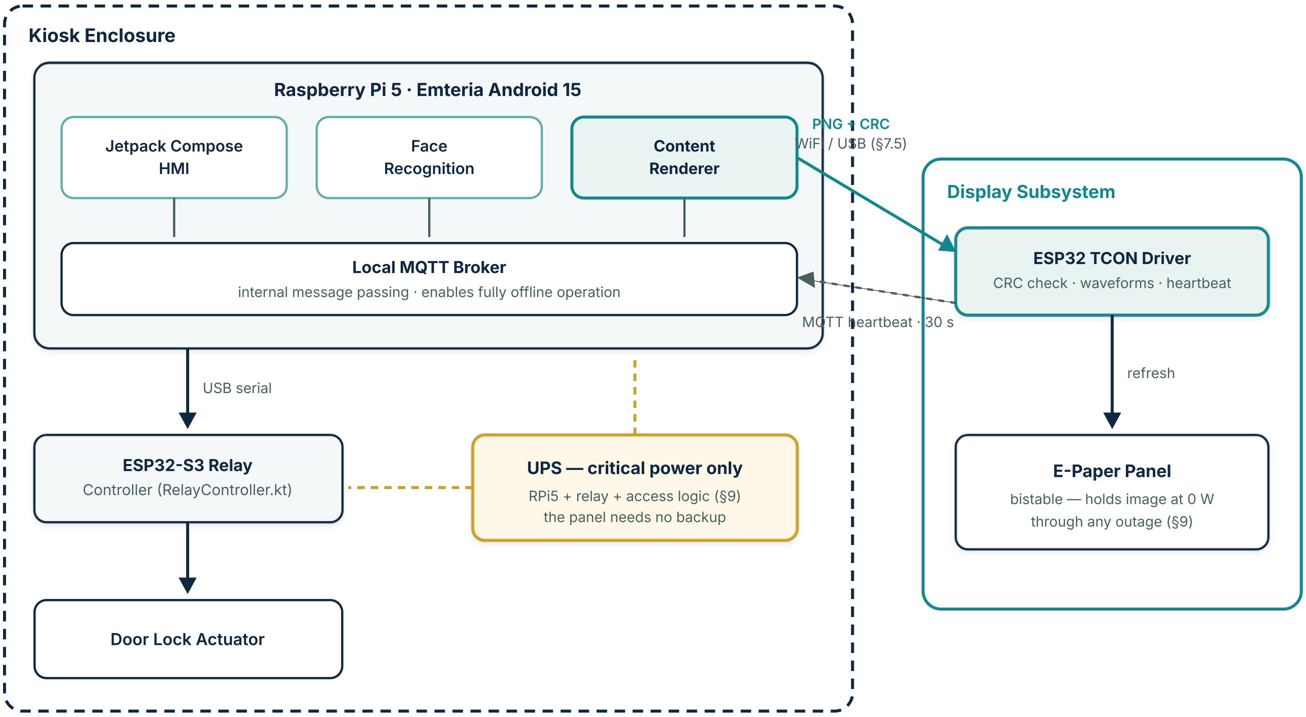


Figure 10 — Full kiosk system architecture (from the project TDD): the §7 callout as a system view — access-control HMI, content rendering, and panel driving in separate layers; the UPS backs only the critical path, because the bistable panel keeps its image at 0 W.

Application patterns — five ways the image gets onto the panel

Choosing the *update channel* is as architectural a decision as choosing the panel itself. Five proven patterns cover the practical spectrum, from fully manual to fully automatic.

PATTERN	TYPICAL PROJECT	HOW THE IMAGE ARRIVES	HARDWARE EXAMPLE	CHOOSE WHEN
① **Removable media** (USB / TF card)	Shop or lobby advertising poster	Staff export PNG/JPG at panel resolution to a flash stick; the board auto-imports on insert and plays a slideshow	DMPH315E62 finished billboard; GDEP253C02 + Android TCON board	Updates are occasional, staff are on-site, and no network is wanted
② **Phone push** (Bluetooth / WiFi)	Restaurant menu, info board	Phone app over BLE (works with no network at all), or a browser upload to the board's built-in web server on the same LAN	GDEP133C02 + ESP32-133C02 (web server included)	Daily price/content edits from a phone, no PC in the loop
③ **NFC tap** (battery-free)	Door labels, shelf tags, name badges	A 2–5 second tap delivers *both* harvested power and the image over the same 13.56 MHz field (§10)	Waveshare NFC modules 1.54"–7.5"; ST25R3911B reader for bulk programming	Small B&W labels with zero wiring and zero batteries

PATTERN	TYPICAL PROJECT	HOW THE IMAGE ARRIVES	HARDWARE EXAMPLE	CHOOSE WHEN
4 **Cloud CMS** (4G / internet)	City-wide outdoor signage fleet	A central CMS schedules content per site and pushes over 4G; units return battery/temperature telemetry; solar + battery power	Geniatech EPC2530-class units with remote CMS	Many remote displays needing central control and health monitoring
5 **Self-generating host**	Live dashboard — and this report's kiosk	The host renders PNG from data sources (APIs, schedules) on a timer and pushes it through the transport abstraction; the in-image timestamp guards freshness	RPi5 + ESP32 TCON — exactly Figures 1–2	Content derives from data; zero-touch operation

Patterns 1→5 trade manual effort for autonomy. The kiosk project is pattern 5; patterns 1–4 are reusable recipes for sibling deployments (ads, menus, labels, fleets) on the same panel knowledge in this report.

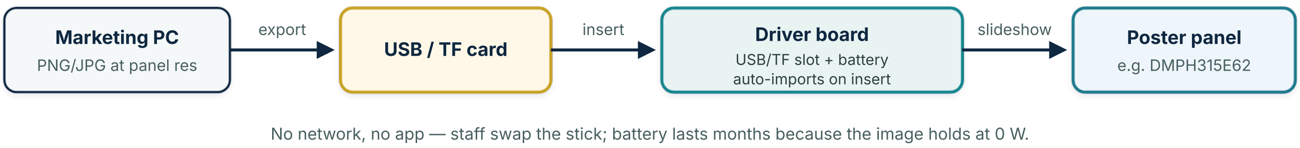


Figure 5 — Pattern 1: shop/lobby advertising poster updated by removable media.

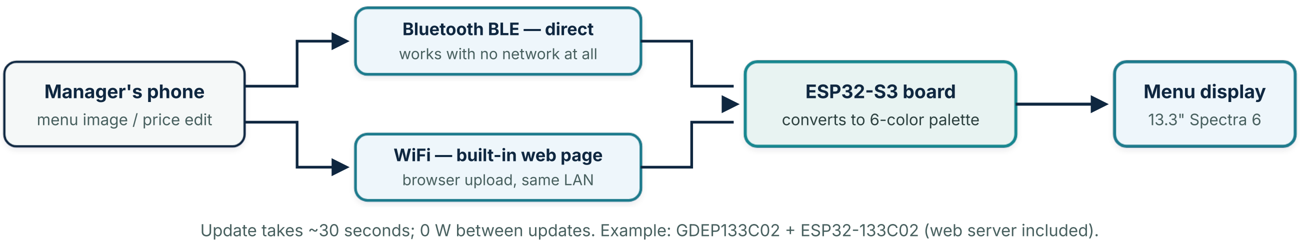
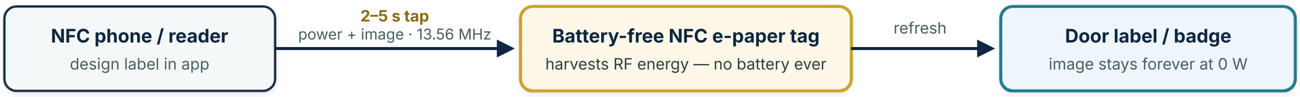
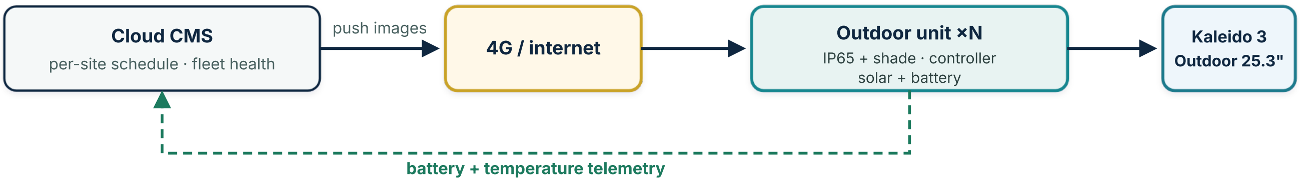


Figure 6 — Pattern 2: restaurant menu updated from a phone over Bluetooth or WiFi.



Sizes 1.54"–7.5", B&W (§10). Bulk programming: ST25R3911B reader board — works with all phones' tags.

Figure 7 — Pattern ③: battery-free door labels and badges updated by an NFC tap.



Solar easily sustains it — content shows at 0 W between updates; telemetry catches overheating and stale units.

Figure 8 — Pattern ④: city-wide outdoor fleet managed remotely from a central CMS over 4G.

Zero-touch: nobody uploads anything — content updates itself. This is the kiosk architecture of Figures 1–2 and 4.

Figure 9 — Pattern ⑤: live information dashboard (reception/mosque) with fully automatic image generation.

7.5 Content-push protocol — formal specification (from the project TDD)

The handshake of Figures 2 and 4 is pinned down in the project's technical design document as an implementable wire protocol. The Android renderer pushes the image over HTTP to the ESP32 TCON; status and heartbeats return over the local MQTT broker already running on the RPi5 — no cloud involved.

Image push (Android → TCON, HTTP):

```
POST /display/update
Content-Type: application/octet-stream
X-Image-CRC: <CRC-32 hex>
X-Image-Width: <pixels>
X-Image-Height: <pixels>

<PNG binary data>
```

RESPONSE	MEANING
`200 OK`	Image received, CRC validated, refresh initiated.
`400 Bad Request`	CRC mismatch or malformed image — the app must re-render and re-push, never retry the same bytes blindly.
`503 Service Unavailable`	TCON busy with a previous refresh (tens of seconds on large panels, slower in heat §12) — back off and retry.

Refresh-complete event (TCON → broker, MQTT `display/status`):


```
{ "event": "refresh_complete",
  "duration_ms": 2400,
  "panel_temp_c": 42 }
```


Heartbeat (TCON → broker, MQTT `display/heartbeat`, every 30 s):


```
{ "timestamp": "<ISO-8601>",
  "uptime_s": 86400,
  "last_refresh_crc": "A1B2C3D4" }
```


Staleness rules: heartbeat interval 30 s; alarm on **3 consecutive missed heartbeats (90 s)** — raise the TCON-stale alarm, log the event, attempt reconnection (the Figure 4 alarm branch). Two useful by-products of this payload design: `panel_temp_c` doubles as the §12 thermal telemetry, and `last_refresh_crc` lets the app verify *what is actually on glass*, not just what was last sent.

SECTION 08

Indicative Price Table (BOM) by Size

08

Estimates to guide budgeting; confirm with the supplier (panel and board are usually separate).

SIZE	TECHNOLOGY	BARE PANEL	MATCHING DRIVER BOARD	COMPLETE KIT
4"	Spectra 6	~$49.52 (Waveshare)	Adapter/HAT ~$10	~$60
7.3"	Spectra 6	~$149 (E Ink)	DESPI-C73 ~$10–15	~$120–160
13.3"	Spectra 6	~$326 (Waveshare) / $449 (E Ink)	ESP32-133C02 ~$40–70	~$298 (Waveshare bundle)
25.3"	Spectra 6	~$1,400 (E Ink retail)	Factory TCON (required)	~$1,000 (ODM)–1,400
31.5"	Spectra 6	~$580–600 (Good Display)	ESP32-S3 board (required)	~$700–1,200
13.3" (outdoor)	Kaleido 3 Outdoor	~$449 (E Ink)	CONCERTO (separate)	On request
25.3" (outdoor)	Kaleido 3 Outdoor	On request	Luvia (separate)	On request
31.2" (grayscale)	Carta B&W, −15→65°C	~$580 (GDEP312TT3)	IT8951 ~$40–70	IP65 unit (DMPH312EC1 / V32-OSM) on request

Reading the columns: bare-panel figures are typically E Ink *retail*, while complete-kit figures are *ODM* (Good Display / Waveshare) — so an ODM kit can legitimately cost less than E Ink's retail bare panel. This is a cross-supplier difference, not a pricing error; confirm each figure and package contents with the supplier. Add an IP65 enclosure & shading to outdoor budgets, and ~5% customs duty on imports into Qatar.

SECTION 09

Power-Outage Behavior & Image Retention

09

All of the displays in this report are bistable electronic paper: the pigment particles physically hold their position once set, so the panel draws power **only while the image is changing** and **zero power to keep an image on screen**. There is no backlight to lose. This makes e-paper uniquely suited to power outages and off-grid sites — the opposite of LCD/LED/OLED, which go black the instant power is removed.

Key Questions Answered

- **Does the image stay visible when power is cut?** Yes, on every display here — the last image freezes and remains fully readable with zero power, **indefinitely** (months to years).
- **Can it operate during an outage?** To *show* the last image: always. To *keep updating* content: a power source (battery or solar) is required, because a refresh needs power.
- **How long without electricity?** Displaying the frozen image: effectively unlimited. Continued updates: battery/solar-dependent (see table).

DISPLAY	LAST IMAGE STAYS W/O POWER?	POWER SOURCE	KEEPS UPDATING OFF-GRID FOR...
Samsung EM32DX (32" reference)	Yes — indefinitely	Built-in 4,600mAh + USB-C	~200 days @ 1 update/day; up to ~500 days if rarer
Good Display 31.5" billboard (DMPH315E62)	Yes	Built-in lithium battery	Untethered; scales with update frequency
Panel + ESP32-S3 board	Yes	Battery header (add LiPo)	Months at low update rates
Waveshare 13.3" + ESP32-S3	Yes	Onboard lithium-battery header	Depends on your LiPo
Bare panel only (any size)	Yes	None (you supply)	Image persists forever; updating needs your power
Kaleido 3 Outdoor (25.3"/13.3")	Yes	Solar-capable (external)	Long-term on solar — depends on panel/battery sizing & cleaning (Qatar dust)
Geniatech EPC2530 (outdoor 25.3")	Yes	External battery option	Extended off-grid with battery

DISPLAY	LAST IMAGE STAYS W/O POWER?	POWER SOURCE	KEEPS UPDATING OFF-GRID FOR...
NFC battery-free e-paper tags	Yes	None — harvests RF from the reader	No stored energy at all; image persists indefinitely (smaller sizes)
Marquee / ChLCD / Carta	Yes (all bistable)	Per your integration	Per power source

- **"0.00W" is the static figure** — technically under 0.005W, reported as 0.00W per IEC standards. A refresh briefly draws a few watts, and battery runtimes assume a fixed update cadence (Samsung's 200 days = one update/day; optional battery kits on AC models last roughly 3–6 weeks at 4 updates/day). Frequent updates shorten runtime sharply.
- **Critical for the kiosk:** the panel retaining its image during an outage does *not* mean the system keeps working. The e-paper keeps showing the last screen, but the controller (RPi5 / ESP32) and any door/access logic still need power to read a card, respond, or push a new image. Use e-paper to keep *information* visible through an outage; give interactive/control functions their own UPS or battery.
- **Heat shortens battery life (not image retention):** in Doha's heat, lithium-battery runtime degrades faster, so battery-powered units may fall short of their rated days — but the bistable image itself is unaffected and still holds at zero power. Solar, NFC, or wired options sidestep this.
- **UPS sizing is currently unsizeable:** the peak power draw of a large-panel refresh (25.3"/31.5") is not published — request it as an RFQ line item or measure it on the 13.3" prototype before specifying the kiosk UPS (§11).

BOTTOM LINE: for power-unreliable contexts, e-paper is ideal — whatever was last displayed stays readable for the entire outage at no power, and battery/solar units can even keep refreshing.

Sources: *Samsung (EM32DX user manual, Newsroom), E Ink, Good Display, Geniatech.*

SECTION 10

NFC E-Paper: Battery-Free, Phone/Reader-Updated Displays

10

NFC e-paper is a distinct branch that needs **no battery and no separate driver board**. The NFC chip on the module *is* the driver, and it harvests its power wirelessly from whatever updates it — there is no SPI host, no TCON, and no wiring. You tap an NFC phone (or hold an NFC reader) near the module; the same RF field both powers the chip and transfers the new image.

Where it sits in the driver taxonomy (§6): unlike the SPI adapters (DESPI), the color TCONs (ESP32-133C02 / Luvia), or the grayscale IT8951, an NFC module is fully integrated — the "driver board" and the "host" collapse into (a) an on-module NFC+EPD chip and (b) an external NFC field source (phone or reader).

10.1 How it works

Bring an NFC phone/reader within a few centimetres of the module's antenna → the RF field powers the chip and carries the image → the refresh finishes in a few seconds → the image then stays with **zero power, indefinitely**.

10.2 Two typical workflows

- **Update the image from your phone's gallery:** in the module's Android app, pick a photo (or shoot one), crop/resize/rotate, adjust brightness and contrast, then tap the phone to the module to push it.
- **An e-paper name/business card updated by tap:** the same flow, but the content is a card design (name, title, logo, QR) or a text image made in the app. Card-like sizes: 2.13" (250×122) or 2.9".
  Important distinction: here NFC is the *update* channel (you tap to change what is shown). If instead you want a *visitor* to tap the card and save your contact (vCard), that is a different function requiring a standard NDEF tag — do not conflate the two.

10.3 Available NFC e-paper (Waveshare)

Sizes: 1.54", 2.13", 2.7", 2.9", 4.2", up to **7.5" (800×480)**. Black & white (some small 3-color), low resolution, few-second refresh. Intended for price/asset/shelf labels (ESL), name badges, and small info tags; all keep the last image with no power.

10.4 NFC devices compatible with e-paper

DEVICE TYPE	EXAMPLE	NOTES
NFC smartphone	Android + the module's app	Easiest; powers + transfers on tap. The vendor app excludes some phones (Samsung/Google/Sony) and iPhone access is limited. Fine for small sizes.
Dedicated NFC reader board	Waveshare ST25R3911B NFC Board / Kit	The matched reader — marketed as ideal for refreshing passive NFC e-papers; ~1.4W output, onboard MCU, standalone or host-integrated. Recommended for larger sizes (4.2"/7.5") and for automation.
High-power HF reader IC/module	ST25R3916, TI TRF7970A, NXP PN5180	Workable if RF output is high enough and the protocol matches (NFC Forum Type 2/4/5, ISO 14443/15693). Verify per module.
USB NFC reader (PC)	Desktop reader/writer	Batch programming from a computer; must be high-power and protocol-matched.
Industrial NFC writer / handheld	ESL encoders	For programming many labels in retail/asset deployments.

Key requirement

The reader must deliver **enough RF output power** to both harvest-power and drive the EPD (the ST25R3911B's ~1.4W is sized for this) — low-power tag readers may fail, especially on larger panels. The module's protocol/commands must also match, so the vendor app or the matched reader is the guaranteed path.

FIT FOR THIS PROJECT: NFC e-paper maxes out around **7.5" and is B&W** — harvested power cannot refresh a large or color panel, so it is **not** an option for the 25"/31.5" color signage. Its place is small **battery-free auxiliary labels** (asset tags, room/desk labels, name badges) around the kiosk, and as the zero-stored-energy extreme of the retention behavior in §9.

Sources: *Waveshare (NFC e-paper module wikis; ST25R3911B NFC Board), STMicroelectronics (ST25R3911B).*

SECTION 11

Procurement Roadmap: Requirements First, Prototype, then Purchase

11

The flagship options in Tracks 1–2 should not be ordered straight from the tables: the outdoor flagship is quote-only, the deciding requirements are usually unwritten, and the large color panels are expensive enough that sequencing matters. This section turns the tracks into a purchase plan.

11.1 Step 1 — write the one-page content spec

One page decides the central trade-off, and it must exist before any panel ≥25.3" is ordered. It states: **content type** (menu / wayfinding / status / promotional), **update cadence** (per hour / day / week), **viewing distance**, and the **install site's measured thermal reality** (not assumed — survey the actual spot, especially glass-fronted interiors per §2). The decision rule it feeds: if cadence is low and the content is informational, the wide-temp B&W **GDEP312TT3** (~$580, −15→65°C) covers *both* tracks with one SKU and halves the budget; the color dual-track stands only if color is a genuine stakeholder requirement (e.g., brand/showcase value), not an inherited assumption.

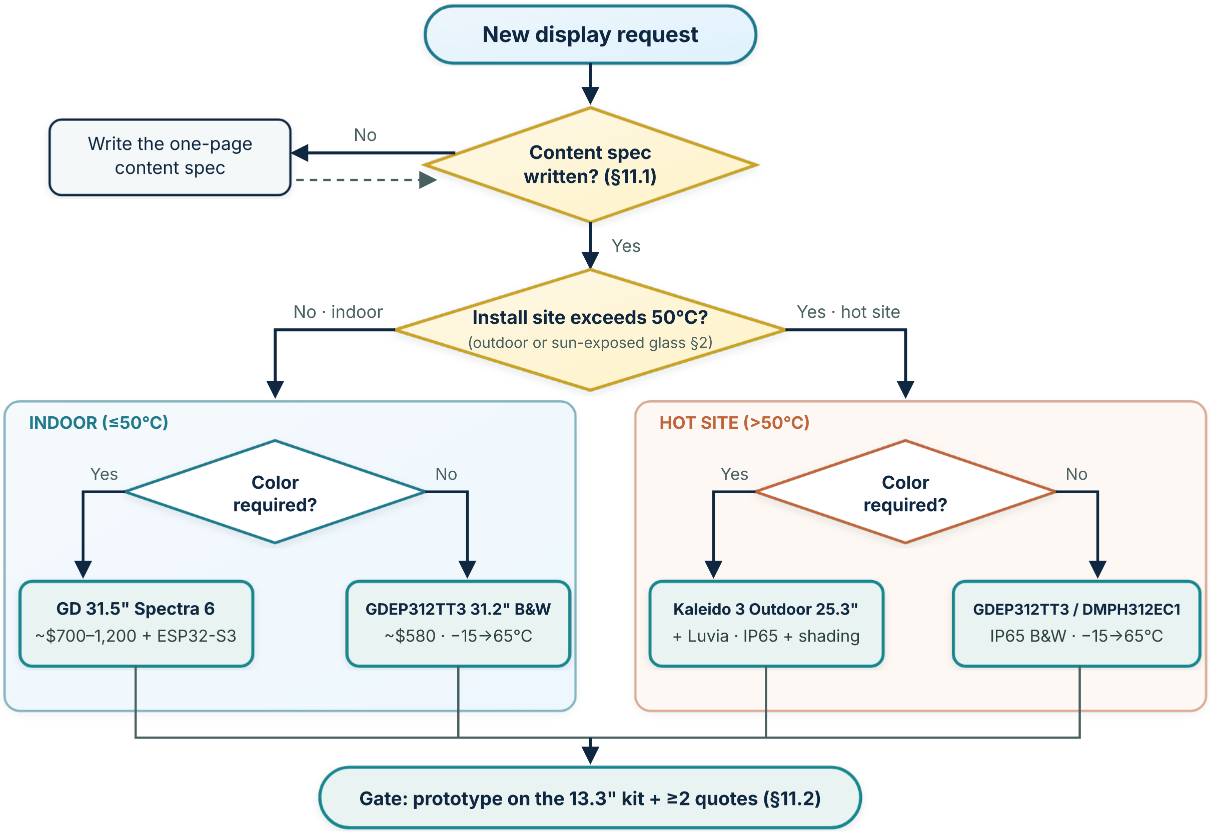


Figure 3 — Panel-selection decision tree: the content spec is the first gate; site temperature and the color question pick among four outcomes, and every path ends at the prototype-and-quotes gate.

11.2 Step 2 — prototype and quotes, in parallel

PHASE	ACTION	COST / TIME	OUTPUT
P0 — now	Sequence diagram of the Android→TCON content push, with link-down, power-outage, and staleness states made explicit	< 1 hour	**Delivered ✓** — Figure 4 below; editable master in the companion pack (Appendix)
P0 — now	Three identical-spec RFQs: **Good Display** (GDEP315C01 + driver board / finished DMPH315E62 / GDEP312TT3), **E Ink** (25.3" Kaleido 3 Outdoor + Luvia), **SEEKINK** (S315E6 / S280EC-O)	1–2 weeks wait	Supplier × price × lead-time matrix
P1 — this month	Waveshare 13.3" Spectra 6 + ESP32-S3 kit: build the *full* pipeline (Android app → transport → ESP32 → panel), measure refresh time and 6-color conversion, practice FPC handling on a cheap panel	~$298	Working pipeline; every line of code transfers to the final panel
P2 — last	Final panel purchase per the content spec + at least two quotes; GDEP312TT3 remains the budget fallback	per quote	Committed order with eyes open

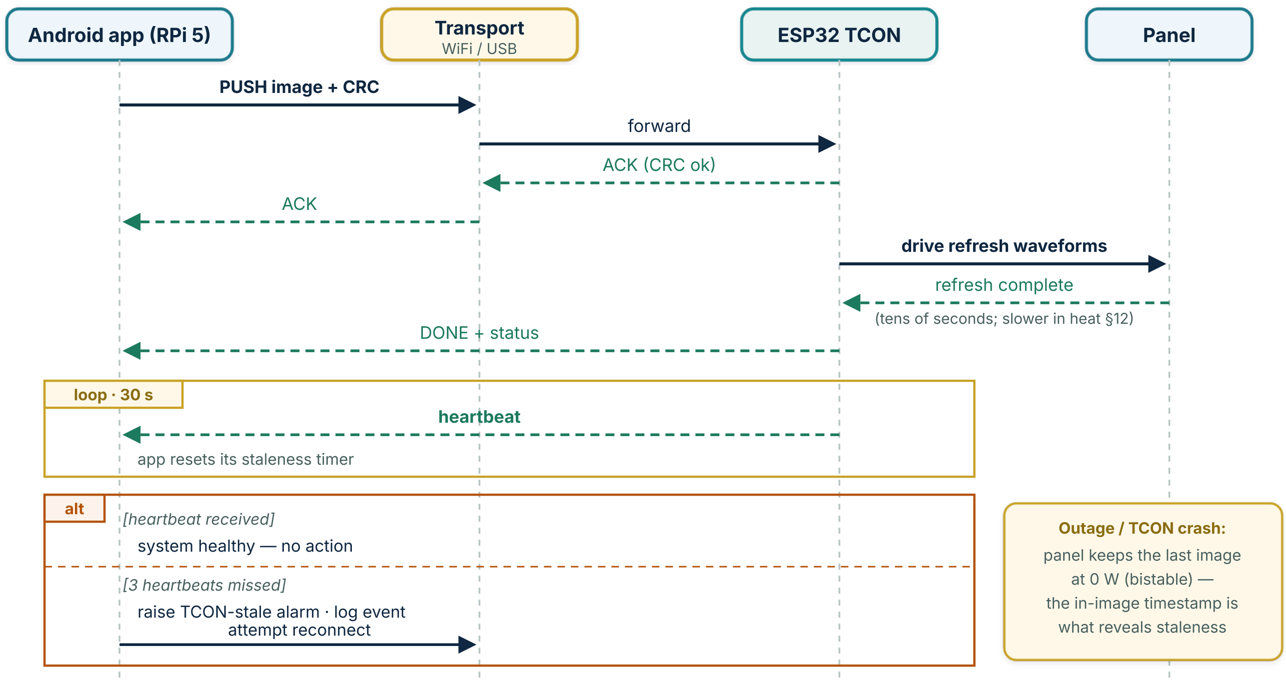


Figure 4 — The delivered P0 artifact: the Android→TCON content-push handshake, the 30-second heartbeat loop, and the 3-missed-heartbeats alarm branch. CRC = integrity checksum appended to each push; ACK = the receiver's confirmation.

Mandatory RFQ line items: unit price · lead time to Doha · MOQ · warranty terms *and DOA policy including return logistics to Qatar* (returning a dead 31.5" panel can cost more than the panel) · Incoterms (request DDP Doha) · **peak refresh power draw** (unpublished; needed for UPS sizing) · and for finished units: **is the content-push API open**, or locked to the vendor app? (An app-locked unit breaks the §7 content interface.)

11.3 Integration hardening checklist

- **Staleness watchdog:** TCON heartbeat + "last updated" timestamp rendered into every pushed image (see §7) — the cheap fix for e-paper's unique failure mode of displaying stale content convincingly.
- **ESD & FPC handling protocol:** Doha's <30% RH indoor air makes ESD a genuine panel-killer. Wrist strap, grounded mat, written insertion procedure; practice on the 13.3" before touching a ~$600 panel.
- **Acceptance test at delivery:** a written checklist run within the DOA window — power-on, full-white, full-black, color bars, ghosting check after 3 refresh cycles. For quote-only B2B imports this is the cheapest insurance available.
- **Content-rotation policy:** alternate at least two images on long-static deployments; static content held for weeks in heat invites ghosting at zero benefit.
- **Site thermal survey:** log the actual install location's temperature across a hot day before final panel selection (feeds Step 1).
- **Hardware qualification (production panels, from the project TDD):** temperature cycling −20→+60°C ×10 cycles in an environmental chamber; 10,000-refresh accelerated life test with no degradation; bench measurement of peak refresh power against the datasheet (this is the number the RFQ asks for and the UPS sizing in §9 is waiting on); water-jet IP65 validation for outdoor variants.

- **Acceptance test — concrete pass criteria (from the deployment SOP):** beyond the visual checks, push a test image and require `200 OK` with a matching `X-Image-CRC`; read `panel_temp_c` from the heartbeat and require agreement with ambient within ±3°C; record refresh power draw and the panel serial + firmware version in the deployment log. **FAIL → quarantine the panel and open the supplier RMA** — this is exactly why §11.2 demands the DOA policy as an RFQ line item.
- **7-day burn-in before handover (from the deployment SOP):** automated alerts on 3 missed heartbeats, panel temperature (>55°C indoor / >65°C outdoor), and UPS state; observe behavior through the 12:00–16:00 peak-heat window; push 3–5 different images across the week and check for ghosting; verify a deliberate TCON power-cut raises the staleness alarm at 90 s and recovers; UPS failover test with ≥15 min runtime; 72-hour fully-offline operation check.

11.4 Phased rollout plan (16 weeks, from the project TDD)

The §11.1–11.3 steps expand into a four-phase schedule in which engineering and procurement run *in parallel* and converge at a week-8 decision gate:

PHASE	WEEKS	ENGINEERING TRACK	PROCUREMENT TRACK	EXIT GATE
1 · Prototype	1–4	Content spec → sequence diagrams → order the 13.3" kit → build the pipeline, measure refresh timing	Identical-spec RFQ package → sent to 3 suppliers → await quotes (1–2 wk)	Working prototype pipeline
2 · Validation	5–8	FPC/ESD handling practice on the cheap panel; complete prototype validation; document lessons	Quotes in → supplier × price × lead-time comparison matrix	**Readiness review: Spec + Quotes + Prototype** — the decision gate
3 · Production order	9–12	—	Apply the Figure-3 decision tree per site → issue the PO → 4–6 wk typical lead → delivery acceptance test (§11.3)	Accepted panel(s) in hand
4 · Deployment	13–16	Site prep + enclosure (1 wk) → panel integration & wiring (2–3 d) → system integration tests (2–3 d) → 7-day burn-in → handover		Operations sign-off

Deployment governance: the field procedure is formalized as **SOP-EPAPER-DEPLOY-001** with a T-day timeline — site assessment & panel-selection verification at T−14…T−7, delivery acceptance test no later than T−7 (*before the panel ever travels to site*), installation at T-day under ESD discipline, integration testing at T+1, burn-in T+2…T+7, final acceptance and Operations handover at T+8 with a complete documentation package (site checklist, acceptance forms, wiring photographs, serials/firmware, burn-in summary).

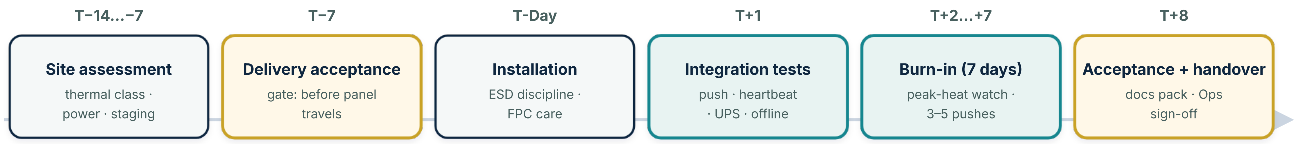


Figure 11 — Deployment governance timeline (SOP-EPAPER-DEPLOY-001): the acceptance test is a hard gate at T−7 — no panel travels to site untested; burn-in alerting (3 missed heartbeats, over-temperature, UPS state) runs through the full week before handover.

11.5 Procurement-tracking data model (the §11.2 RFQ discipline as a schema)

The Appendix's procurement-tracking **ERD** gives the RFQ workflow a relational backbone — six entities chained `suppliers → rfqs → quotes → purchase_orders`, with `quotes` referencing `panels` and panels linked to their compatible `driver_boards` (the §2.2 pairing rule, enforced as a foreign key):

ENTITY	PURPOSE	DECISION-CRITICAL FIELDS
`suppliers`	Vendor registry	`region`, `channel` (direct/distributor), `support_quality`
`panels`	Panel catalog	`technology`, `temp_min_c`/`temp_max_c` (drives the Figure-3 tree), `interface` (drives the board pairing)
`driver_boards`	Matched TCON/adapter per panel	`interface`, `panel_id` (FK — no orphan boards)
`rfqs`	One row per supplier enquiry	`spec_version` — quotes are only comparable against the same content spec
`quotes`	The §11.2 mandatory line items, as columns	`unit_price_usd`, `lead_time_days`, `moq`, `warranty_months`, `doa_policy`, `incoterms`, **`peak_refresh_power_w`**, **`push_api_open`**
`purchase_orders`	Commitment & landed cost	`customs_duty_pct` (§12: ~5% into Qatar), `landed_cost_usd`, `acceptance_test_passed`

The two bold columns are the report's signature requirements made queryable: a supplier who cannot fill `peak_refresh_power_w` blocks the §9 UPS sizing, and `push_api_open = false` flags a vendor-locked unit that breaks the §7 content interface. Open items carried in the TDD: confirm the actual Qatar customs duty rate with Finance, IP65 availability of GDEP312TT3 from the preferred supplier, and MTBF targets for the acceptance criteria.

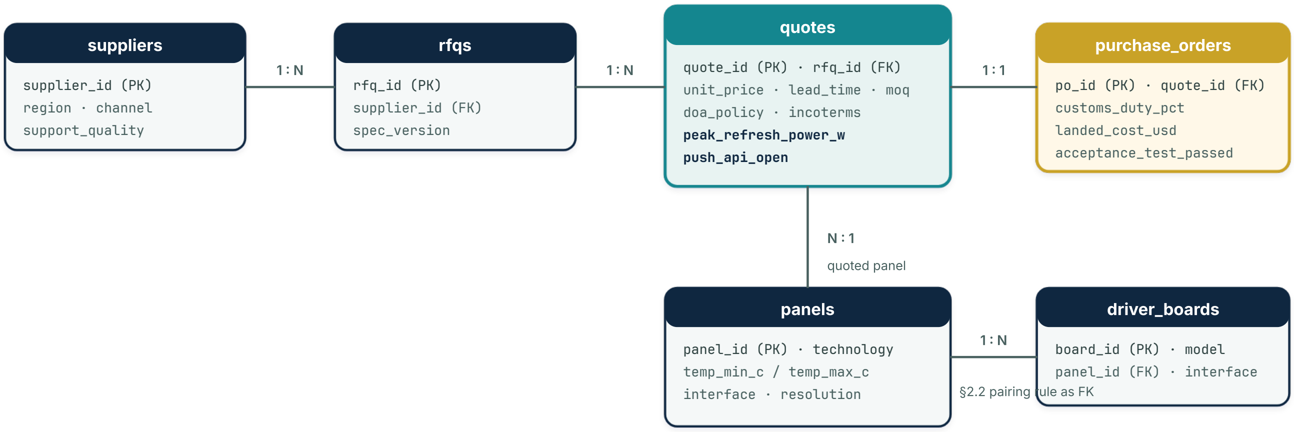


Figure 12 — Procurement-tracking ERD (from the project TDD): the §11.2 RFQ discipline as a schema; bold fields in `quotes` are the two report-signature requirements — missing `peak_refresh_power_w` blocks UPS sizing, and `push_api_open = false` flags a vendor-locked unit.

SECTION 12

Engineering Considerations for Qatar

12

- **Enclosure & shading (outdoor):** even −15/65°C panels need an IP65 enclosure + sun-shade + anti-glare; direct Doha sun can push the surface above 65°C.
- **Do not mix technologies:** Spectra 6 = indoor only; Kaleido 3 Outdoor / Marquee / ChLCD = outdoor.
- **Refresh speed:** slows in high heat — account for it in your content cadence.
- **ESD (dry climate):** Qatar's low indoor humidity (<30% RH) raises ESD risk during FPC/panel handling — ground straps and a written handling protocol are mandatory at the integration bench (§11.3).
- **Ghosting prevention:** rotate content (≥2 alternating images) on long-static deployments; verify ghosting in the delivery acceptance test (§11.3).
- **Power:** 0W static suits solar/battery installation with no permanent power cable.
- **Customs:** ~5% customs duty on imports (no VAT yet) — budget for it in the landed cost.
- **Sourcing channels:** Good Display via good-display.com / buyepaper.com / Alibaba; E Ink directly via shopkits.eink.com; Kaleido Outdoor and Marquee by contacting E Ink or its partners; ChLCD via IRIS Optronics; small panels and HATs via Waveshare/AliExpress/Amazon.

Security considerations (from the project TDD)

- **Physical:** intrusion sensors on the kiosk enclosure; ESD discipline at every FPC touch; IP65 sealing verified on outdoor installs (gasket check is an SOP step, not an assumption).
- **Network:** the kiosk WiFi AP lives on an isolated VLAN; the system is fully offline by design (TDD non-goal NG2), so there is no cloud attack surface; admin access requires physical presence + authentication.
- **Data integrity:** CRC-32 on every image transfer before any refresh; heartbeat monitoring bounds TCON-failure detection at 90 s; every content update is audit-logged with timestamp and checksum.
- **Supply chain:** supplier vetting recorded in the §11.5 tracking model; 100% delivery acceptance testing; ESP32 TCON firmware signed and verified at boot.

SECTION 13

Final Recommendations (Scenario Recipes)

13

SCENARIO	SUGGESTED RECIPE	CONTROLLER / CONNECTION
Indoor color signage/menu ~32"	Good Display 31.5" Spectra 6 + ESP32-S3 board	ESP32 (WiFi) or Android over WiFi
Cheap color prototype	Waveshare 13.3" E6 + ESP32-S3	ESP32 / RPi
Integration with your own MCU (small)	Bare panel ≤13.3" + DESPI	RPi (40-pin) / STM32 (SPI)
Outdoor color signage (Doha)	Kaleido 3 Outdoor 25.3" + Luvia + IP65 enclosure	E Ink board + shading
Outdoor, max color robustness/sun	ChLCD (IRIS Optronics)	B2B solution
Economical outdoor (color not needed)	GDEP312TT3 (−15/65°C), or IP65 V32-OSM / DMPH312EC1	IT8951 / built-in (USB/WiFi)
Your kiosk (RPi5 + Android + ESP32)	Panel + TCON as controller; RPi5/Android as content source	WiFi/USB to the TCON

SECTION 14

Reference: Benchmark Display & Market Context

14

Benchmark display — Samsung Color E-Paper (EM32DX): 32", QHD 2560×1440, based on E Ink Spectra 6, weighs 2.5 kg (with battery), 17.9 mm thick, 0.00W static consumption (per IEC62301: anything below 0.005W is reported as 0.00W), launched June 2025 at ~$1,080. Every alternative in this report is also 0W static because it is genuine bistable e-paper.

Market context: E Ink Holdings is the world's largest ePaper display supplier (around 55–60% of the market), and most Chinese manufacturers are converters that depend on it (the notable exception is OED, with its own film). Spectra 6 won SID's Display of the Year award, and E Ink is building a new fab (H6) in Taoyuan at roughly $107 million to scale up large-format panels.

This comprehensive report is based on official product pages and public sources as of June 2026. Prices are estimates and subject to change; confirm price, package contents (panel and board are usually separate), and temperature range directly with the supplier before purchasing.

Document History (internal revisions v7–v10, June 2026)

v10 delivers three of the Appendix's external diagram masters as native in-report figures, drawn in the report's own visual language directly from the companion documents: **Figure 10** (§7) — the full kiosk system architecture from TDD §3.1, showing the HMI / face-recognition / content-renderer layer over the local MQTT broker, the ESP32-S3 relay → door-lock chain, the display subsystem, and the UPS scoped to the critical path only; **Figure 11** (§11.4) — the SOP-EPAPER-DEPLOY-001 T-day governance timeline with the T−7 acceptance-test hard gate; and **Figure 12** (§11.5) — the procurement-tracking ERD with the two report-signature RFQ fields highlighted. Appendix B updated to mark the delivered masters. All v9 content and features are preserved.

v9 folds the project's two companion engineering documents into the report, turning architectural guidance into implementable specification: **§2.3** design goals & non-goals (G1–G6 / NG1–NG4) from the TDD; **§7.5** the content-push wire protocol made formal (HTTP `POST /display/update` with CRC headers, 200/400/503 semantics, MQTT `display/status` and `display/heartbeat` payloads, 30 s / 3-missed staleness rules); **§11.3** extended with the hardware qualification matrix (temperature cycling, 10,000-cycle life test, peak-power bench measurement, IP65 water-jet), concrete acceptance pass criteria (`200 OK` + CRC match, ±3°C temperature-sensor check, RMA-on-fail), and the 7-day burn-in protocol; **§11.4** the 16-week phased rollout with the week-8 readiness gate plus the SOP-EPAPER-DEPLOY-001 T-day governance timeline; **§11.5** the procurement-tracking data model whose `quotes` entity encodes the mandatory RFQ line items as queryable columns; and a **§12 security block** (physical / network / data-integrity / supply-chain). Glossary extended with MQTT, RMA, Burn-in, and MTBF. All v8 content and features are preserved.

v7–v8 are presentation and accessibility passes with all technical content preserved: interactive navigation with section tracking, a searchable 30+ term technical glossary with inline tooltips, dark mode, expandable sections, enhanced figures, full A4 print/PDF output (collapsed sections auto-expanded, repeated table headers, page-break control), responsive layout, and keyboard/touch accessibility throughout.

Appendix — Companion Diagram Pack (editable masters)

Every figure and process in this report is maintained as a live, editable master diagram in the project's internal workspace (Lucidchart and Eraser). All ten masters were visually verified on the live canvas — layout and arrow semantics — before publication; the editable files are available from the author (Eng. Abdulaziz N. AL-Haddad) on request.

A. Integration figures (= Figures 1–2, §7): the physical signal chain, and the kiosk content-push architecture.

B. Engineering & procurement set: ① *panel-selection decision tree* (= Figure 3, §11.1); ② *content-push sequence diagram* with the heartbeat loop and alarm branch (= Figure 4 — the delivered §11.2 P0 artifact); ③ *full kiosk system architecture* — **now delivered in-report as Figure 10** (§7); ④ *procurement-tracking ERD* — **now delivered in-report as Figure 12** (§11.5); ⑤ *three-lane procurement BPMN* (§11.2); ⑥ *deployment T-day timeline* — **delivered as Figure 11** (§11.4, from the SOP).

C. Application patterns set (= Figures 5–9, §7): ① USB/TF advertising poster; ② phone-updated restaurant menu (BLE / WiFi web server); ③ battery-free NFC labels; ④ city-wide 4G/CMS outdoor fleet with solar and telemetry; ⑤ self-generating live dashboard (the kiosk pattern).

D. Companion engineering documents (Eraser workspace): ① *Technical Design Document — E-Paper Display System for Doha Access Control Kiosks* (goals/non-goals, layered architecture, content-push protocol spec, procurement data model, testing strategy, 16-week rollout) — integrated into this report as §2.3, §7.5, §11.3–11.5 and the §12 security block; ② *SOP-EPAPER-DEPLOY-001 — E-Paper Display Kiosk Deployment* (roles, T-day timeline, delivery acceptance procedure, installation/ESD discipline, integration tests, burn-in, handover forms).

APPENDIX

Technical Glossary

A

Alphabetical reference of the 35 technical terms used throughout this report.

ACK — Acknowledgment signal - confirms successful receipt of data in the content-push handshake protocol.

Bistable — The property of e-paper where each pixel maintains its state without power - the image stays visible at 0W.

BOM — Bill of Materials - a comprehensive list of components, parts, and prices needed for a project.

Burn-in — A supervised initial operating period (7 days in the deployment SOP) with automated monitoring, used to surface early failures before handover to operations.

CMS — Content Management System - software for scheduling and pushing content to digital signage fleets remotely.

CRC — Cyclic Redundancy Check - an error-detecting code used to verify data integrity during image transfers.

DDP — Delivered Duty Paid - an Incoterm where the seller bears all costs and risks until the goods are delivered to the buyer.

DESPI — Good Display's SPI adapter board series for small e-paper panels (C02, C73, C1085, etc.).

DOA — Dead on Arrival - a product that is non-functional when first received. Critical for return policies on expensive panels.

EPD — Electrophoretic Display - the technical name for e-paper technology using charged pigment particles in fluid.

ESD — Electrostatic Discharge - a sudden flow of electricity that can damage sensitive electronic components like FPC connectors.

ESL — Electronic Shelf Label - small e-paper tags used in retail for dynamic pricing, often battery-free with NFC.

FPC — Flexible Printed Circuit - a thin, flexible ribbon cable that connects the e-paper panel to the driver board. Fragile and must be handled with care.

Framebuffer — A memory buffer storing the image to be displayed - large color panels require significant RAM (e.g., ESP32 over Arduino).

Ghosting — A faint residual image from previous content that remains visible after refresh - caused by incomplete particle movement.

Heartbeat — A periodic signal from the TCON to the host confirming the system is alive - used by the staleness watchdog.

Incoterms — International Commercial Terms - standardized trade terms defining buyer/seller responsibilities in global shipping.

IP65 — Ingress Protection rating: dust-tight and protected against water jets - essential for outdoor enclosures in Qatar.

IT8951 — A generic driver IC for grayscale Carta e-paper panels, supporting USB, SPI, and I80 interfaces.

Level Shifter — A circuit that converts logic levels (e.g., 5V Arduino to 3.3V panel) to prevent damage to the display.

LVDS — Low-Voltage Differential Signaling - a high-speed interface used by the 25.3" Kaleido 3 Outdoor panel.

MOQ — Minimum Order Quantity - the smallest number of units a supplier will sell in a single order.

MQTT — Message Queuing Telemetry Transport - a lightweight publish/subscribe messaging protocol. The kiosk runs a local broker on the RPi5 for heartbeats and status events, fully offline.

MTBF — Mean Time Between Failures - a reliability metric; target values feed the acceptance test criteria (open question in the TDD).

NDEF — NFC Data Exchange Format - a standardized format for storing data on NFC tags (e.g., vCards, URLs).

PMIC — Power Management Integrated Circuit - regulates the multiple voltage rails required by e-paper panels.

QSPI — Quad SPI - a faster variant of SPI using four data lines, required by large panels like the 31.5" Spectra 6.

RF — Radio Frequency - electromagnetic waves used by NFC to power and communicate with battery-free e-paper tags.

RH — Relative Humidity - Qatar's low indoor RH (<30%) increases ESD risk during panel handling.

RMA — Return Merchandise Authorization - the supplier's formal authorization to return a defective unit. Agree RMA/DOA logistics to Qatar in the RFQ before ordering large panels.

SPI — Serial Peripheral Interface - a synchronous serial communication protocol used by small e-paper panels (up to 13.3").

Staleness Watchdog — A monitoring mechanism that detects when the TCON stops updating, preventing outdated content from being displayed indefinitely.

TCON — Timing Controller - the chip on the driver board that generates the precise waveforms needed to update the e-paper display.

UPS — Uninterruptible Power Supply - a battery backup that keeps critical systems running during power outages.

Waveform — The specific voltage sequence applied to e-paper particles to move them between color states - unique per panel type.